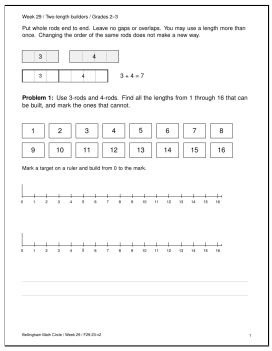
Week 29 / Two-length builders / Problem menu

for adults: pick, then print



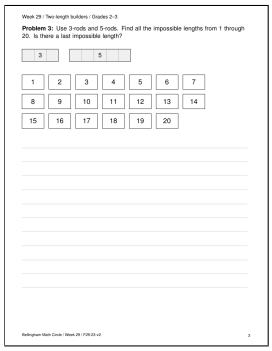
Which lengths 3-rods and 4-rods build
K–1 P1 · Grades 2–3 P1 · Grades 4–5 P1
first route
wording differs by band



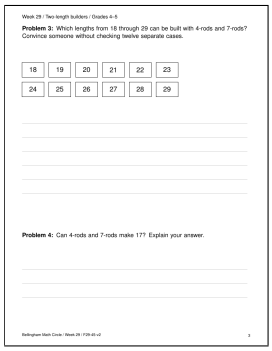
Every way to build 12, 15, 16 and 24 with 3-rods and 4-rods
K–1 P2 · Grades 2–3 P2
first route
wording differs by band



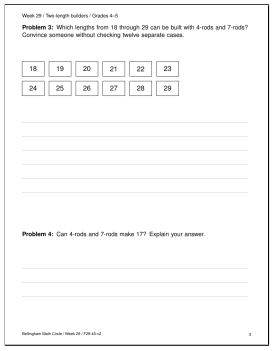
Last impossible lengths for 3 and 5, 4 and 5, 4 and 7
Grades 4–5 P2
first route



Which lengths 3-rods and 5-rods build
K–1 P3 · Grades 2–3 P3
first route
wording differs by band



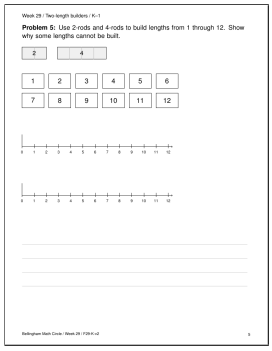
Lengths 18 through 29 with 4-rods and 7-rods
Grades 4–5 P3
return route



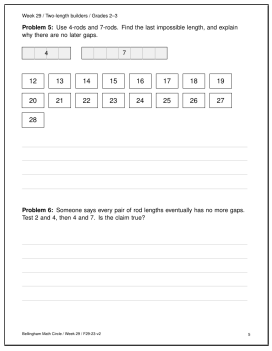
Can 4-rods and 7-rods make 17?
Grades 4–5 P4
return route



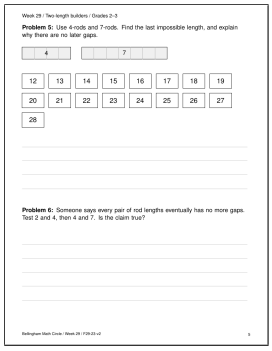
Every length from some point on
K–1 P4 · Grades 2–3 P4
return route
wording differs by band



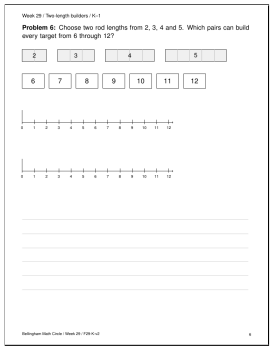
Lengths 1 through 12 with 2-rods and 4-rods
K–1 P5
return route



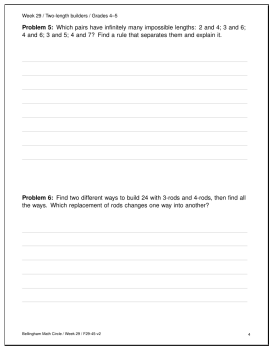
The last impossible length for 4-rods and 7-rods
Grades 2–3 P5
return route



Does every pair eventually have no more gaps?
Grades 2–3 P6
return route



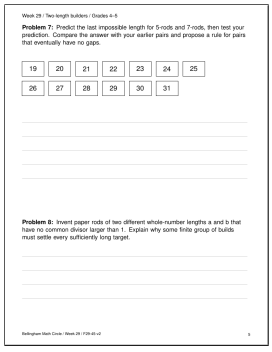
Which pairs from 2, 3, 4 and 5 build every target 6 through 12
K–1 P6
return route



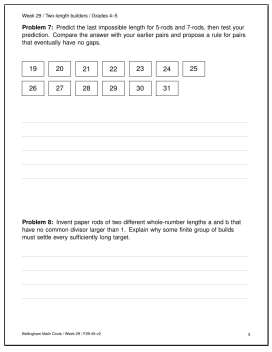
Which pairs have infinitely many impossible lengths
Grades 4–5 P5
return route



All the ways to build 24, and the exchange between them
Grades 4–5 P6
return route



Predict and test the last impossible length for 5 and 7
Grades 4–5 P7
return route



Invented rods with no common divisor
Grades 4–5 P8
return route